

SERGIO CRISTANCHO

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Languages: Spanish (native), English, German, Italian.

PERSONAL STATEMENT

I am a graduate student in Applied and Computational Mathematics at Princeton University, advised by June Huh. My interests lie in the intersection of algebra, geometry and combinatorics.

EDUCATION

Princeton University P.h.D. in Applied Mathematics.	Princeton, New Jersey <i>Ongoing</i>
Universidad de los Andes M.S. in Mathematics, B.S. in Physics, B.S. in Mathematics.	Bogotá, Colombia 2023 2021 2020

RESEARCH

Publications:

Harmonic Hierarchies for Polynomial Optimization. With Mauricio Velasco. 2024. *SIAM Journal on Optimization* 2024 34:1, 590-615. <https://doi.org/10.1137/22M1484511>

Preprints:

Extremal Graphs for the Lights Out Problem. With Julien Codsi, Alexander Divoux and Varun Sivashankar. 2026. <https://arxiv.org/abs/2602.07241> (Submitted to journal)

Tree metrics and log-concavity for matroids. With Federico Ardila, Graham Denham, Chris Eur, June Huh and Botong Wang. 2026. <https://arxiv.org/abs/2601.02547> (To appear in Proceedings of the London Mathematical Society)

Talks:

Tree metrics and log-concavity for matroids. *FPSAC 2026*, University of Washington, July 2026

Inequalities for metric trees and matroids. *Algebra Seminar*, University of Kentucky. April 2026.

Lorentzian polynomials and triangular hyperfields. *Arbeitsgemeinschaft: Combinatorial Hodge Theory*, Mathematisches Forschungsinstitut Oberwolfach, October 2025.

Inequalities for metric trees and matroids. *Discrete Math Seminar*, University of Massachusetts Amherst, September 2025.

Three proofs of the reverse Khovanskii-Teissier Inequality. *Special Year Seminar*, Institute for Advanced Study, January 2025.

Harmonic Hierarchies for Polynomial Optimization on Homogeneous Spaces. *SIAM Conference on Applied Algebraic Geometry 2023 (SIAM AG23)*, Eindhoven University of Technology, July 2023.

Harmonic Hierarchies for Polynomial Optimization. *International Conference on Continuous Optimization (IC-COPT 2022)*, Lehigh University, July 2022.

Harmonic Hierarchies for Polynomial Optimization. *Conferencia Colombiana de Matemáticas Aplicadas e Industriales (MAPI 2)*, June 2022.

TEACHING & COMMUNITY INVOLVEMENT

Teaching Assistant	Princeton University
Abstract Algebra,	2025
Honors Linear Algebra,	2025, 2026
Applied Algebra.	2024
Teaching Assistant	Universidad de los Andes
Real Analysis,	2019
Abstract Algebra,	2019
Enumerative Combinatorics.	2020
Complementary Lecturer	Universidad de los Andes
Linear Algebra,	2019, 2021, 2022
Vector Calculus.	2021
Mentor for the Directed Reading Program	Princeton University
	2024,2025,2026

HONORS AND AWARDS

Fellowships	
Simons Dissertation Fellowship in Mathematics.	2026

INDUSTRY EXPERIENCE

Research and Development	Melius I+D
Mathematical model developer.	Nov. 2020 - Jul. 2023